

Exploring the Impact of Lifestyle on Sleep and Productivity

Checkpoint 1



Data Analysis in Software Engineering 25/26

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Our datasets

Dataset 1

Power Nap vs Coffee Effectiveness

- participant demographics (age, occupation)
- sleep duration
- intervention details (type, duration)
- alertness scores (before and after intervention), along with reported side effects

Dataset 2

Mental Health & Productivity

- daily exercise
- screen time
- diet quality
- stress levels
- mood ratings
- productivity scores

Joining Strategy: K-Nearest Neighbors

For every row in the Power Nap dataset, the single closest row in the Mental Health & Productivity dataset was found in scaled feature space.

Matching features: **sleep_duration**, **mood_rating**, **productivity_rating**

Resulting Dataset

1000 participants

15 features

3 Categorical variables

12 Numerical variables

Data Preprocessing

Missing Values Handling:

side_effects column: NaN values filled with the string "None".
No other significant missing values were found across the dataset.

Data Quality Checks:

Confirmed no duplicate participant_id.
Reviewed distributions and unique counts for key categorical variables to ensure data integrity.

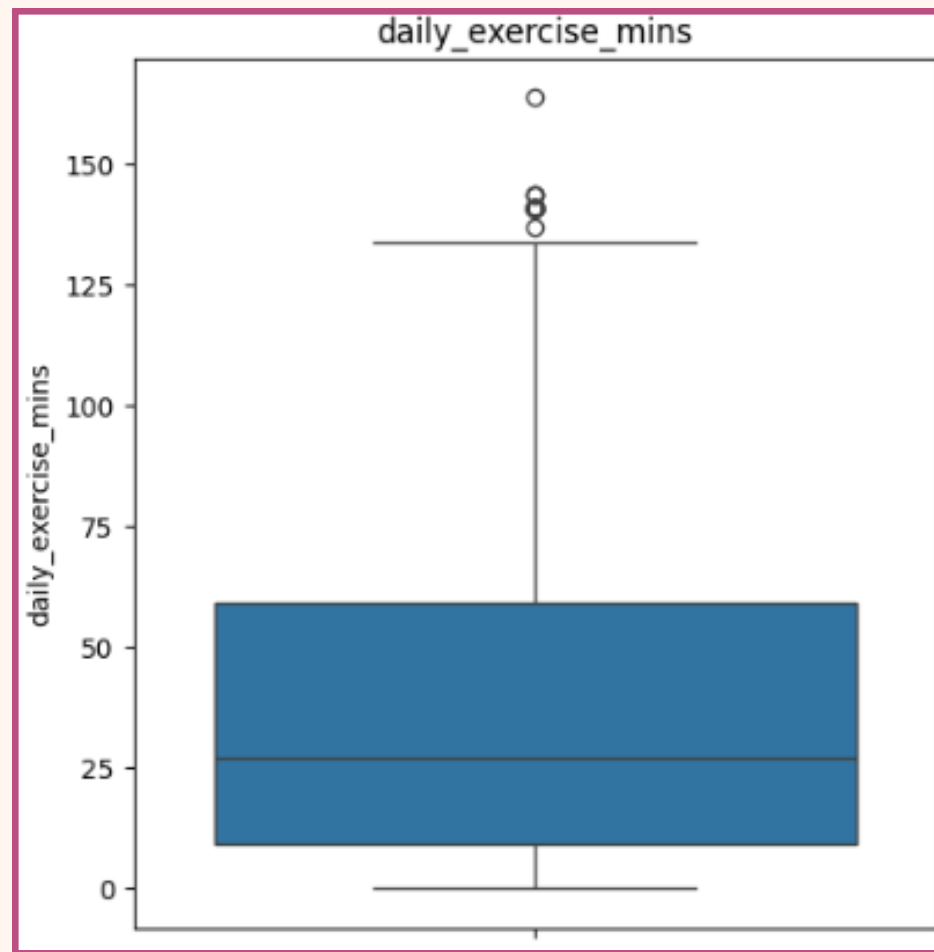
Feature Engineering (Creating New Metrics):

delta_alertness
(alertness_score_after - alertness_score_before)
Measures the absolute change in alertness.

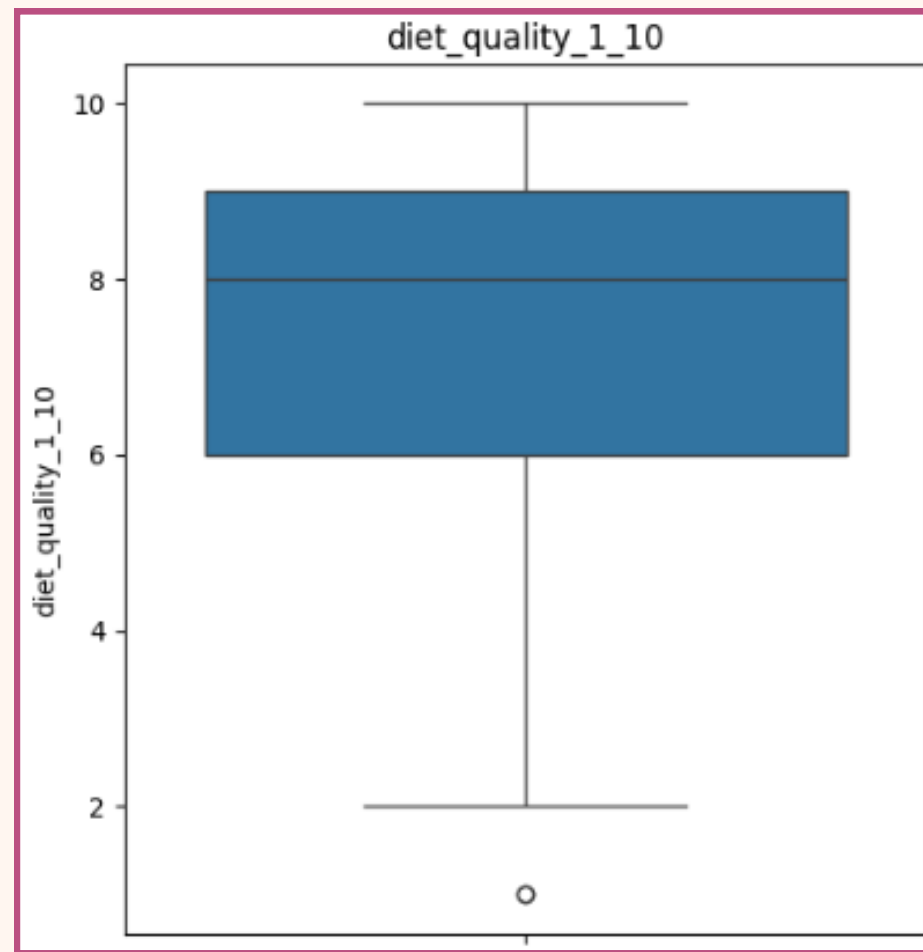
pct_alertness_change
(delta_alertness / alertness_score_before)
Relative measure of change, with care taken to avoid division by zero.

These steps are essential for clean data, creating meaningful metrics for analysis, and ensuring reliability of our findings.

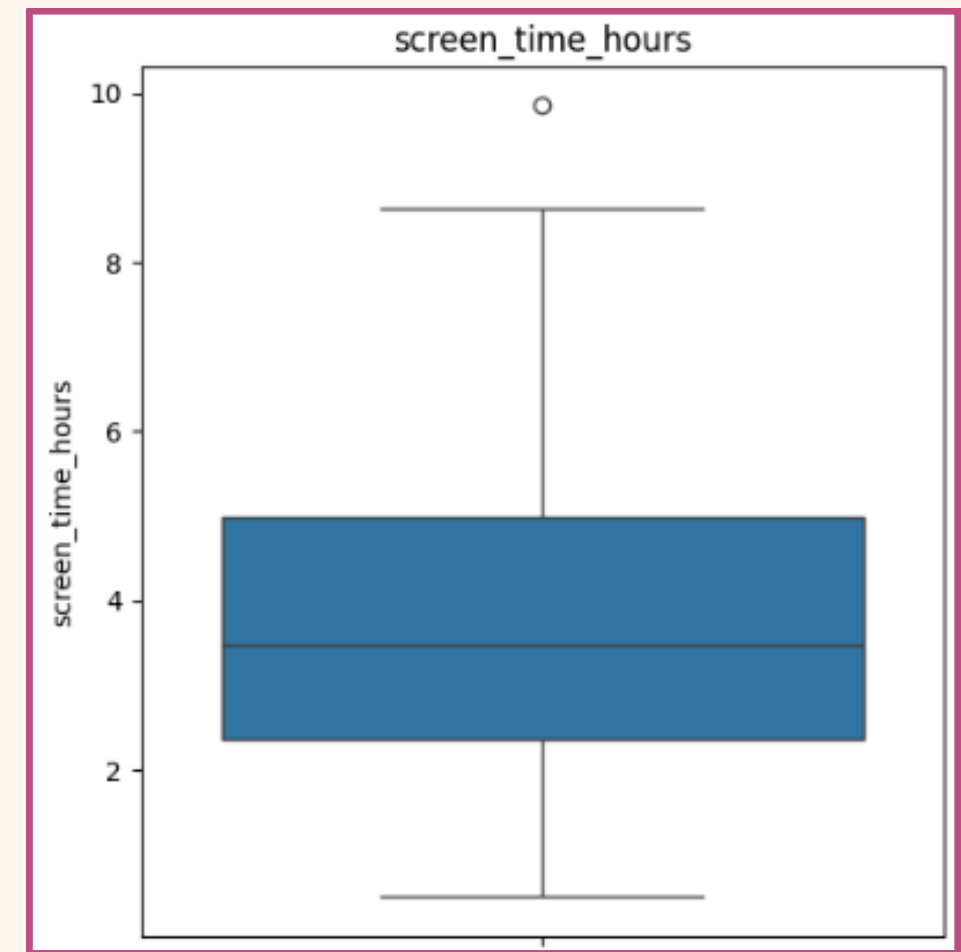
Outliers



59 outliers



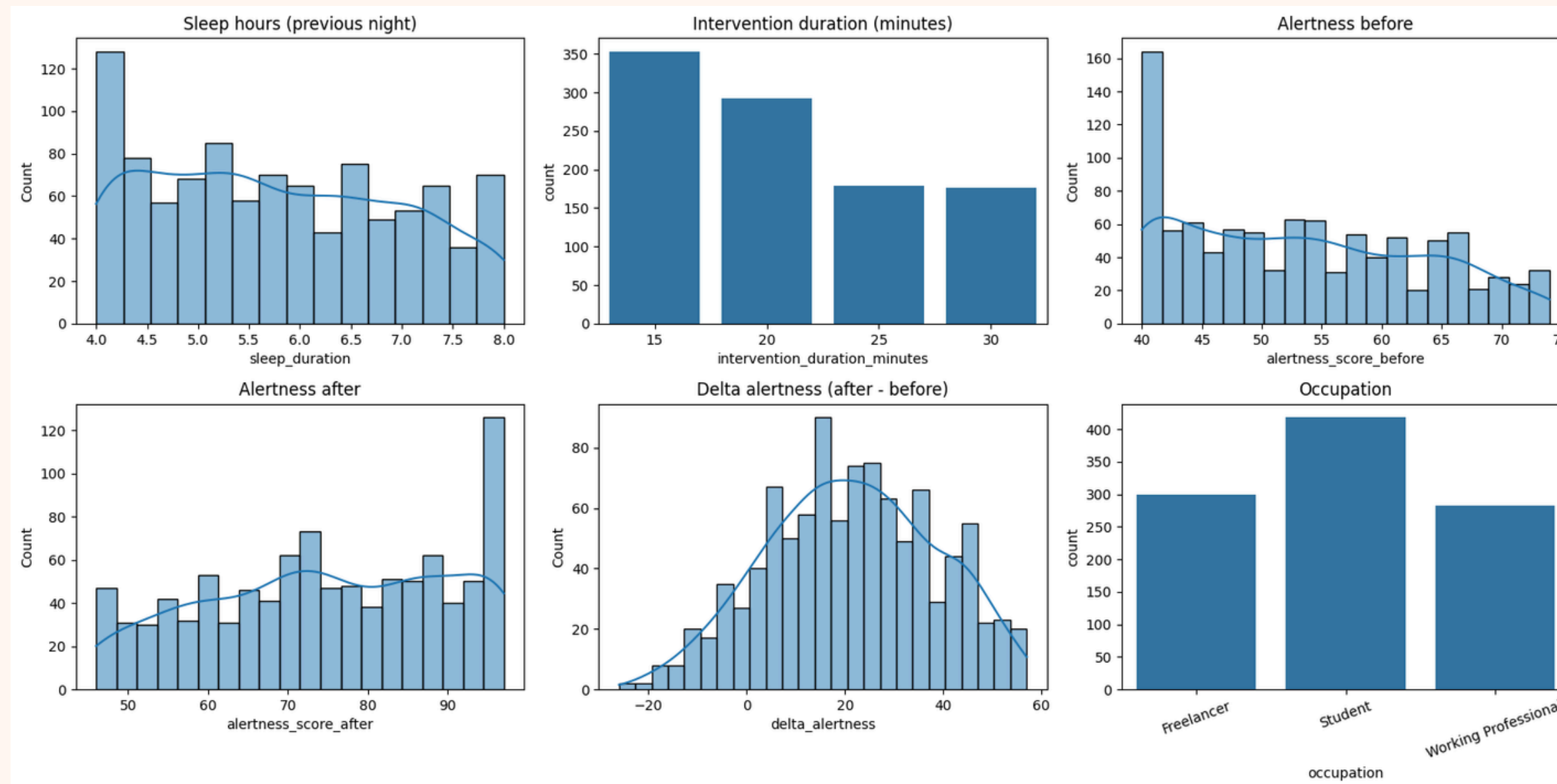
2 outliers



1 outlier

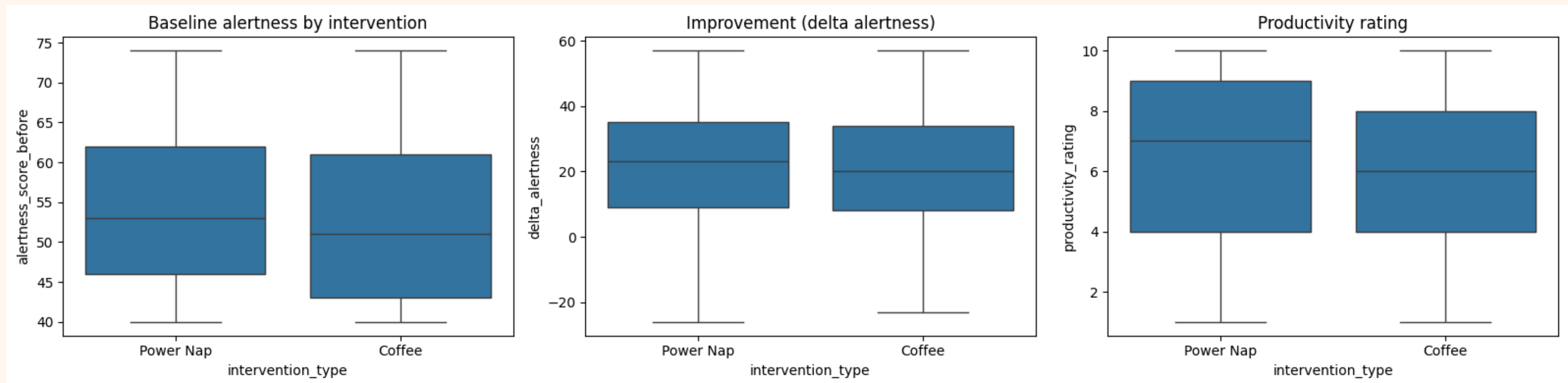
Dataset variability

- Distributions show spread/skew across key variables
- Motivates robust summaries + outlier checks



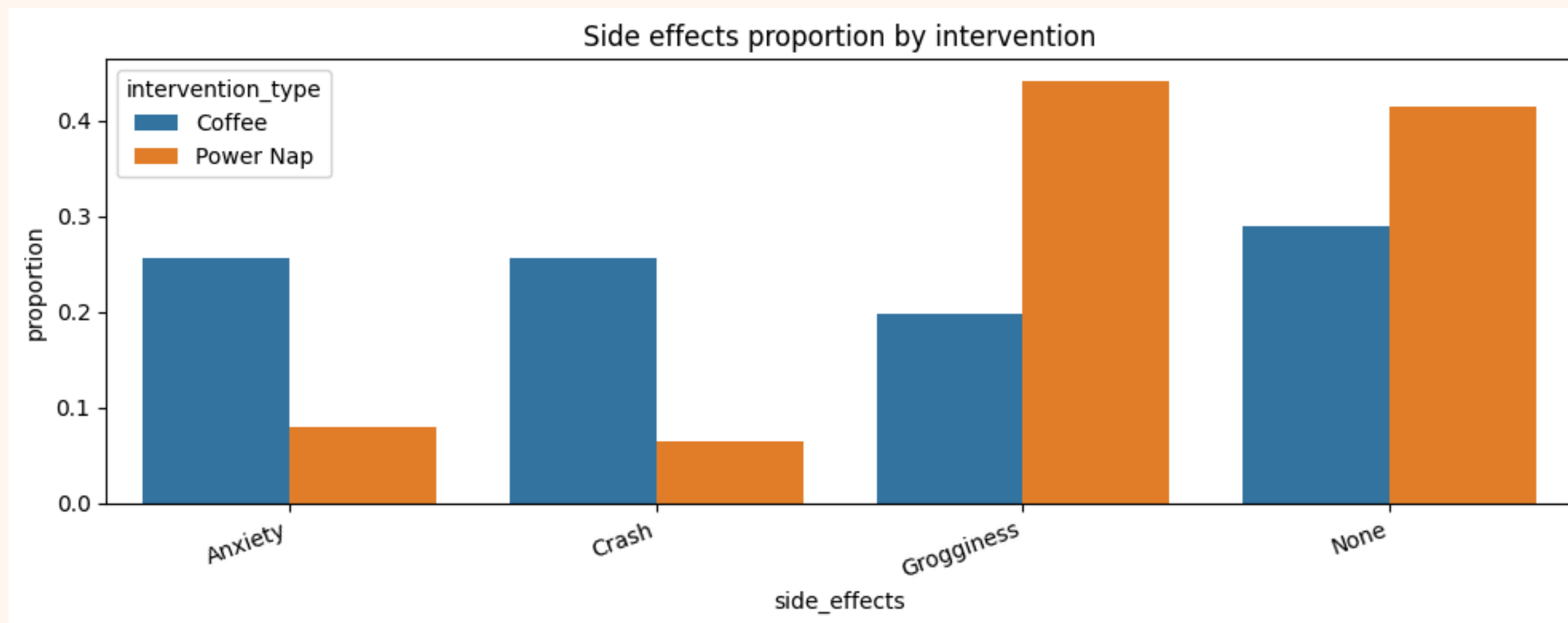
Coffee vs Power Nap: alertness improvement

- We compare interventions using Δ alertness (score after – score before).
- Boxplots let us compare the improvement (median) and consistency for each group.

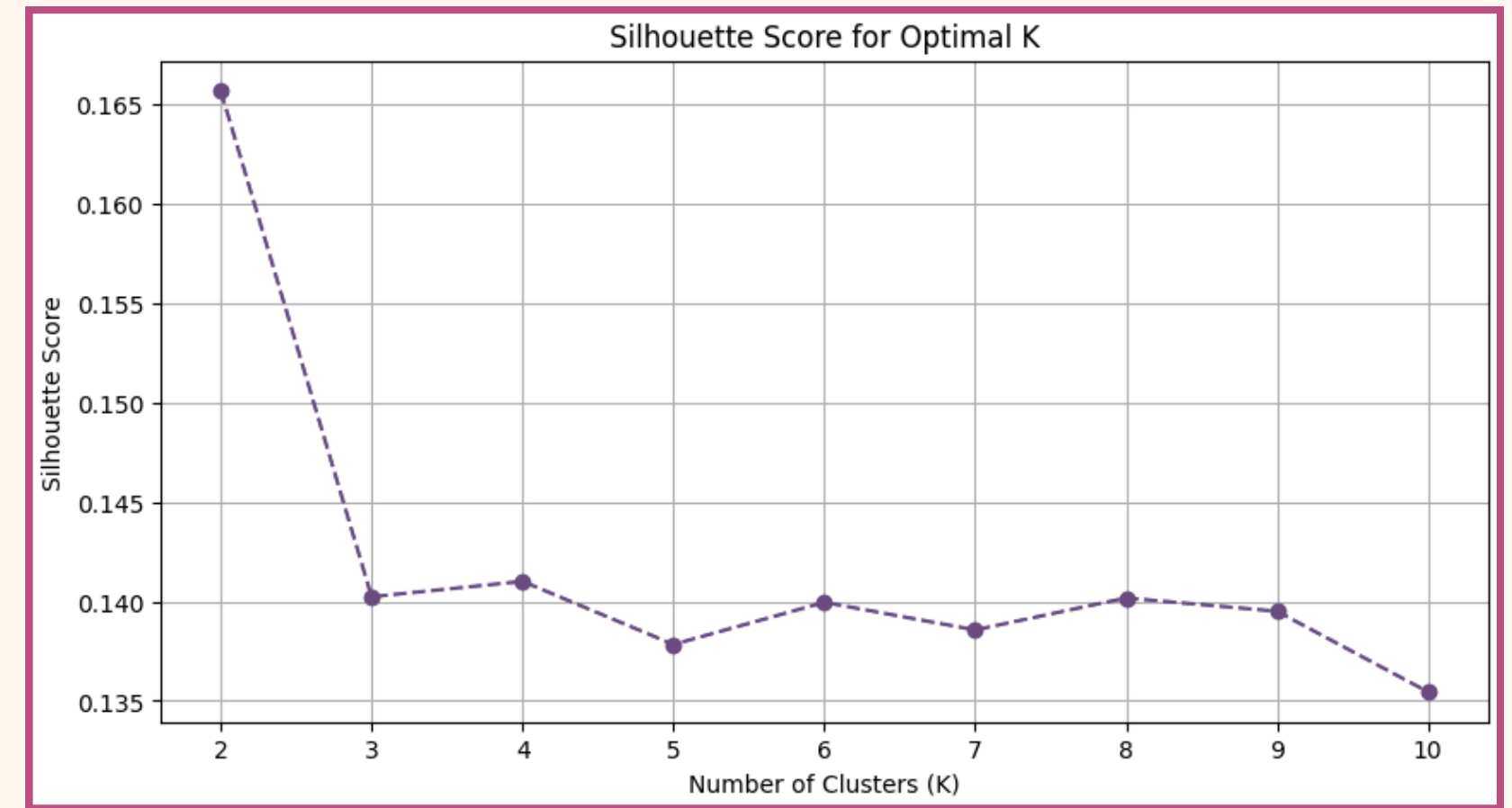
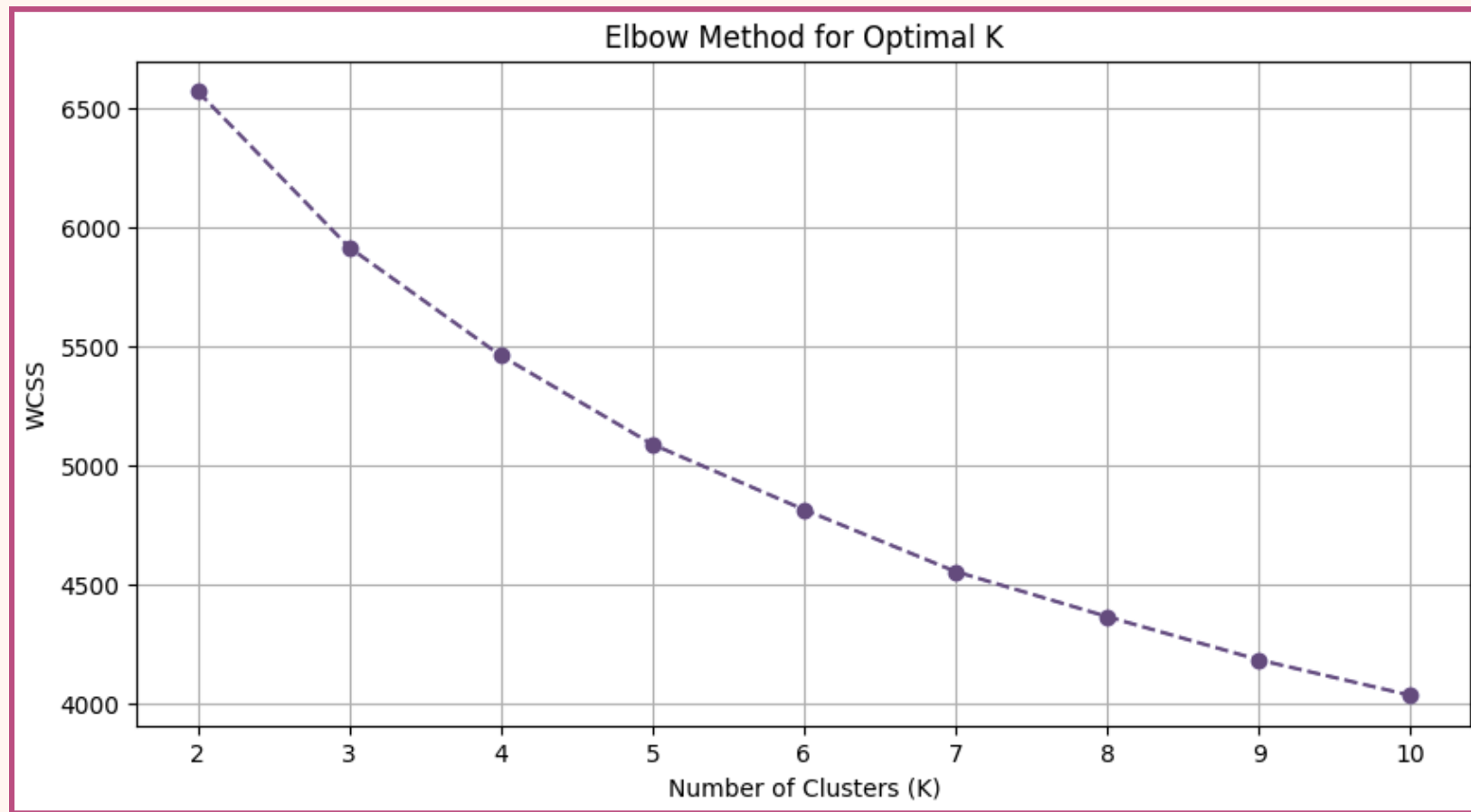


Side effects trade-off by intervention

- Side effects are summarized as proportions within each intervention group.
- An intervention may improve alertness but differ in reported side effects.

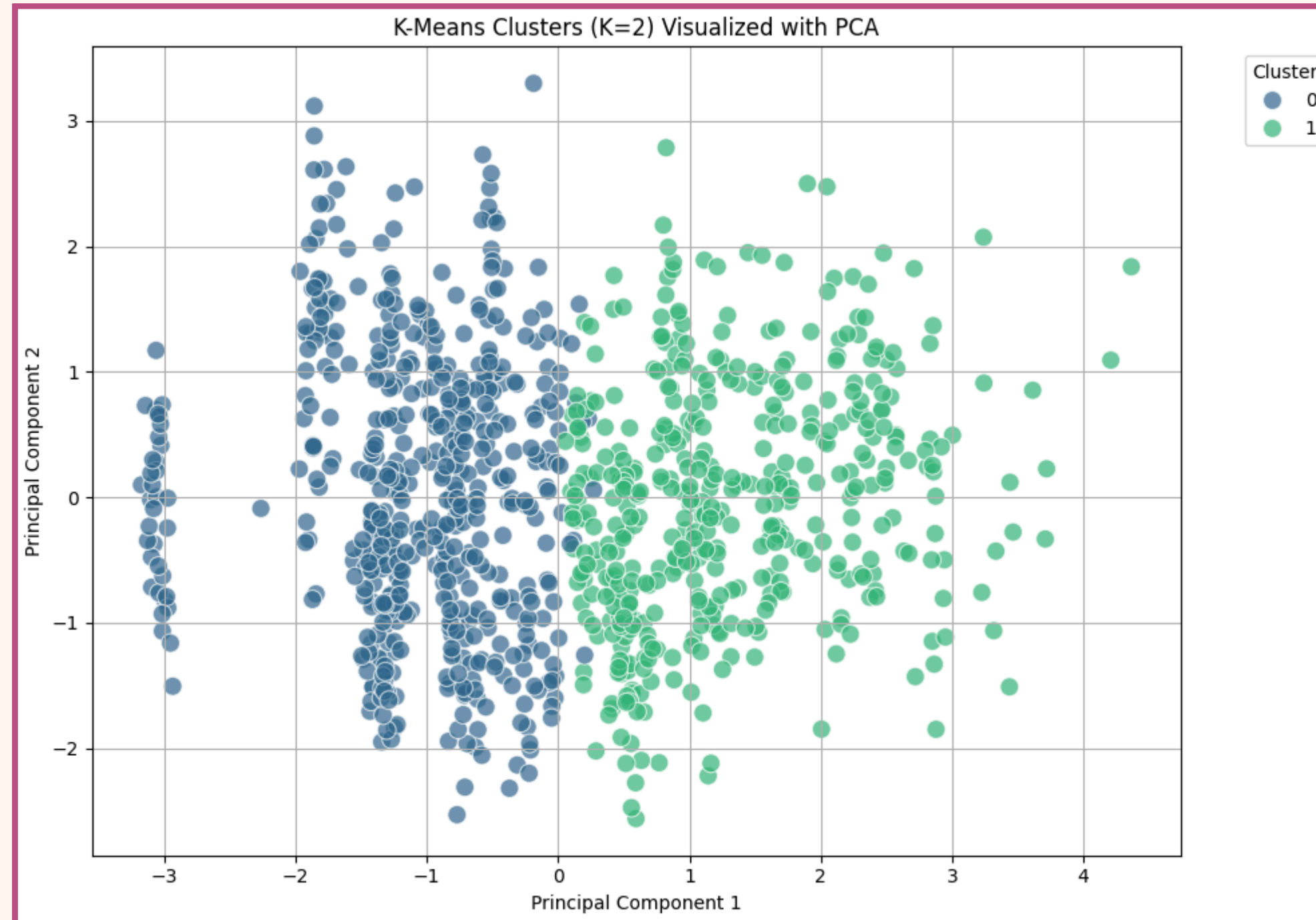


Clustering Analysis - K-Means



Elbow Method and Silhouette Score to find optimal K

Clustering Analysis - K-Means



Clustering Analysis - K-Means

kmeans_cluster	0	1
age	33.73	33.09
sleep_duration	5.73	5.81
intervention_duration_minutes	21.35	20.51
alertness_score_before	53.37	53.47
daily_exercise_mins	19.80	61.54
screen_time_hours	4.49	2.76
diet_quality_1_10	6.26	8.76
stress_level_1_10	7.03	4.31
delta_alertness	20.41	22.07

Persona Profiling

Sedentary Lifestyle VS Active Lifestyle

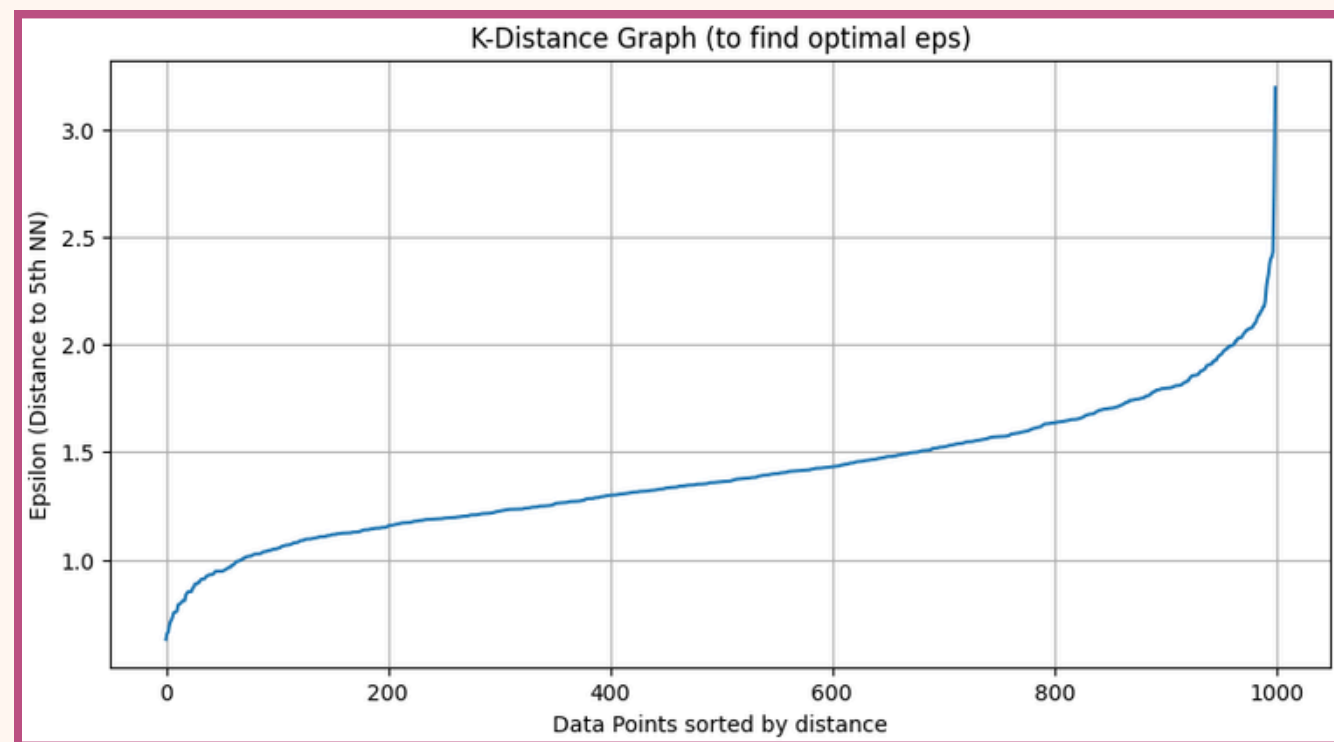
Cluster 0

- **20min** of daily exercise
- **High** screen time
- **Average** quality diet
- **High** stress

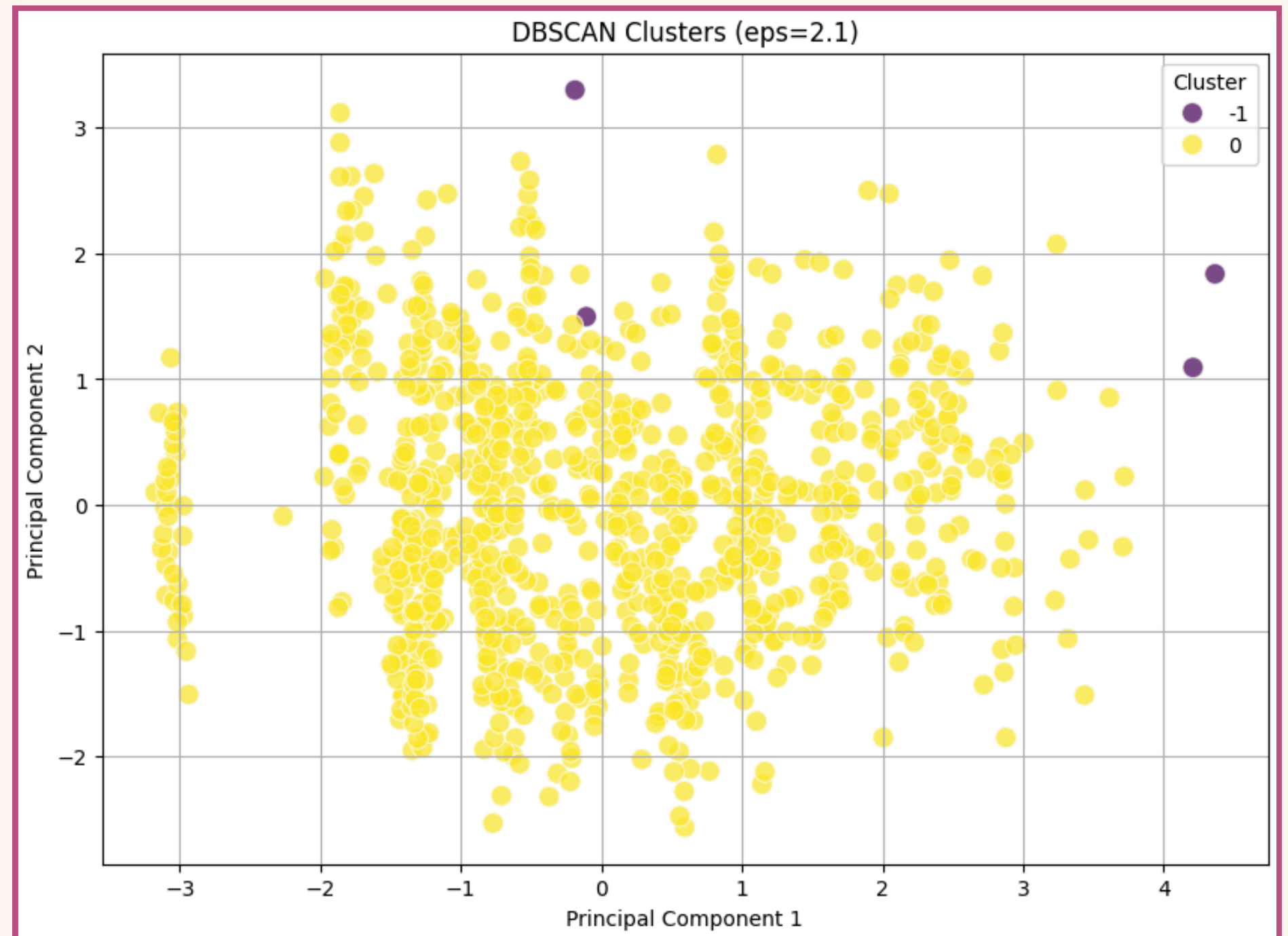
Cluster 1

- **1hr** of daily exercise
- **Low** screen time
- **High** quality diet
- **Low** stress

Clustering Analysis - DBSCAN



Find optimal EPS for DBSCAN cluster



Thank you!

